

# SQL PROJECT

## Exploratory SQL Analysis on Classicmodels Sample Database

By Komal Rathod

## Purpose of the Project:

This project dives into the business data of Classicmodels a fictional retail company. Using SQL we explored sales trends, top-selling products and customer buying patterns. Turn raw data into clear insights that can help improve decisions across sales, inventory and customer service.

## Project Objective:

To explore and analyze business data from the Classicmodels database using SQL, with the goal of uncovering sales trends, customer insights, and product performance.

## Dataset Overview

The data comes from the ClassicModels database, simulating a real-world vehicle retail company. It includes:

- Orders & payments
- Customer & employee info
- Product lines & inventory
- Each table gives a slice of how MotorVista operates, making it a great dataset for hands-on business analysis.

**Tools Used:** My SQL Workbench

# Project Strategy

This is a sample project based on the Classicmodels dataset, which is a fictional business database. The purpose of this analysis is to practice real-world SQL skills and demonstrate how data can support business understanding.

While the data is not from a real company, the analysis simulates how a potential acquisition team might explore customer profiles, sales performance, and operational data to make informed decisions

## Data Analysis Workflow

- Understanding the Dataset
- Defining Business Questions
- Writing SQL Queries
- Extracting Insights
- Presenting the Findings

# ClassicModels Analytics

**By solving key business questions using SQL, I delivered the following types of analysis**

- **Sales Performance Analysis**

Identified top-selling product lines and tracked monthly revenue trends

- **Customer Insights**

Analyzed customer locations, credit limits, and payment histories to identify high-value customers

- **Product Analysis**

Evaluated product line performance, stock levels, and pricing patterns

- **Revenue Analysis**

Monitored total payments received and revenue contribution by customer and region

- **Operational Efficiency Analysis**

Reviewed pending orders, shipment statuses, and employee-office structure for better workflow visibility

## Q.1 . How many employees are in each job title?

```
select jobtitle, count(*) as total_Employees  
from employees group by jobtitle;
```

| jobtitle             | total_Employees |
|----------------------|-----------------|
| President            | 1               |
| VP Sales             | 1               |
| VP Marketing         | 1               |
| Sales Manager (APAC) | 1               |
| Sale Manager (EMEA)  | 1               |
| Sales Manager (NA)   | 1               |
| Sales Rep            | 17              |

## Q.2 What is the total sales revenue?

```
select sum(quantityordered*priceEach)  
as total_sales from orderdetails;
```

| total_sales |
|-------------|
|-------------|

|            |
|------------|
| 9604190.61 |
|------------|

## Q.3 Which 5 customers placed the most orders?

```
select customernumber, count(*) as total_orders
from orders group by customernumber
order by customernumber desc limit 5;
```

|   | customernumber | total_orders |
|---|----------------|--------------|
| ▶ | 496            | 4            |
|   | 495            | 2            |
|   | 489            | 2            |
|   | 487            | 2            |
|   | 486            | 3            |

## Q.4 What is the earliest and latest order date for each customer?

```
select customernumber,min(orderdate) as first_order,  
max(orderdate)last_order from orders  
group by customernumber  
order by customerNumber desc limit 10 ;
```

|  | customernumber | first_order | last_order |
|--|----------------|-------------|------------|
|  | 496            | 2003-07-07  | 2005-04-01 |
|  | 495            | 2003-12-09  | 2004-04-26 |
|  | 489            | 2003-11-14  | 2004-01-22 |
|  | 487            | 2003-09-12  | 2004-02-10 |
|  | 486            | 2003-03-10  | 2004-11-17 |
|  | 484            | 2003-11-14  | 2004-10-06 |
|  | 475            | 2003-12-01  | 2004-01-29 |
|  | 473            | 2003-10-09  | 2004-02-09 |
|  | 471            | 2003-11-21  | 2005-05-09 |
|  | 462            | 2003-10-21  | 2005-03-03 |

## Q.5 How many unique products are there in each product line?

```
select productline, count(distinct productcode) as productcount  
from products group by productline;
```

| productline      | productcount |
|------------------|--------------|
| Classic Cars     | 38           |
| Motorcycles      | 13           |
| Planes           | 12           |
| Ships            | 9            |
| Trains           | 3            |
| Trucks and Buses | 11           |
| Vintage Cars     | 24           |

## Q.6 Join customers and orders to get customer names and their order dates.?

```
SELECT c.customerName, o.orderDate
FROM customers c
JOIN orders o ON c.customerNumber = o.customerNumber;
```

| customerName                 | orderDate  |
|------------------------------|------------|
| Online Diecast Creations Co. | 2003-01-06 |
| Blauer See Auto, Co.         | 2003-01-09 |
| Vitachrome Inc.              | 2003-01-10 |
| Baane Mini Imports           | 2003-01-29 |
| Euro + Shopping Channel      | 2003-01-31 |
| Danish Wholesale Imports     | 2003-02-11 |
| Rovelli Gifts                | 2003-02-17 |
| Land of Toys Inc.            | 2003-02-24 |
| Cruz & Sons Co.              | 2003-03-03 |
| Motor Mint Distributors Inc. | 2003-03-10 |
| AV Stores, Co.               | 2003-03-18 |

## Q.8 Total payment received from each customer – show top 5 payers?

```
SELECT c.customerName, SUM(p.amount) AS total_payment
FROM customers c
JOIN payments p ON c.customerNumber = p.customerNumber
GROUP BY c.customerName
ORDER BY total_payment DESC LIMIT 5;
```

|   | customerName                 | total_payment |
|---|------------------------------|---------------|
| ► | Euro + Shopping Channel      | 715738.98     |
|   | Mini Gifts Distributors Ltd. | 584188.24     |
|   | Australian Collectors, Co.   | 180585.07     |
|   | Muscle Machine Inc           | 177913.95     |
|   | Dragon Souveniers, Ltd.      | 156251.03     |

## Q.7 Find products with a price higher than the average product price?

```
SELECT productName, buyPrice
FROM products
WHERE buyPrice >
      (SELECT AVG(buyPrice)FROM products);
```

|   | productName                          | buyPrice |
|---|--------------------------------------|----------|
| ▶ | 1952 Alpine Renault 1300             | 98.58    |
|   | 1996 Moto Guzzi 1100i                | 68.99    |
|   | 2003 Harley-Davidson Eagle Drag Bike | 91.02    |
|   | 1972 Alfa Romeo GTA                  | 85.68    |
|   | 1962 LanciaA Delta 16V               | 103.42   |
|   | 1968 Ford Mustang                    | 95.34    |
|   | 2001 Ferrari Enzo                    | 95.59    |
|   | 1958 Setra Bus                       | 77.90    |
|   | 2002 Suzuki XREO                     | 66.27    |
|   | 1969 Corvair Monza                   | 89.14    |
|   | 1968 Dodge Charger                   | 75.16    |

# Conclusion

This project gave me the opportunity to approach data like a real business problem. By using SQL to explore the Classicmodels dataset, I was able to answer key questions related to sales, customers, and operations.

I identified top-performing products, high-value customers, revenue trends, and operational gaps such as pending orders and shipping delays.

Most importantly, this project solidifies my SQL skills and reflects my ability to think like a data analyst — turning raw data into clear, actionable insights that support smarter business decisions.



# THANK YOU!

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